

Individual Contribution Report

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Project: Performance Evaluation of Vector Indexing Techniques for AI Search Systems

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1. Role & Responsibilities

I serve as the Vector Indexing & Tuning Lead for this project. My primary responsibility is to design and implement the three indexing strategies — Flat Search, IVF, and HNSW — under identical experimental conditions using FAISS, and to tune their key parameters (IVF nprobe and HNSW efSearch) to map the speed/accuracy trade-off. This component is the core of the benchmark: every recall, memory, and build-time measurement depends on the indexes I construct and on their being built and queried fairly under the same conditions.

Code Modules Owned

- indexing.py — FAISS index wrappers for Flat (exact baseline), IVF (k-means partitioning, nprobe), and HNSW (graph: M / efConstruction / efSearch), with index memory and build-time measurement
- tune_parameters.py — parameter-tuning sweeps for IVF nprobe and HNSW efSearch (speed/accuracy trade-off)

Writing Responsibilities

- Related Work section of the project proposal
- Novelty & Contributions section of the project proposal
- Future Work section of the project proposal
- Index construction / methodology section of the final report

Technical Responsibilities

- Designing a unified index abstraction so Flat, IVF, and HNSW are built and queried under identical conditions
- Configuring FAISS index parameters (IVF nlist / nprobe; HNSW M / efConstruction / efSearch)
- Automatic nlist scaling so IVF clustering is valid at small dataset sizes
- Measuring index memory footprint and index construction time for each method
- Running the parameter-tuning sweeps that map how nprobe and efSearch trade speed for accuracy

2. Weekly Contribution Log

The following log records individual actions and outcomes from Week 1 through Week 8, aligned with the course project timeline and phase milestones.

Week 1 — Course Orientation & Background Reading

Actions Taken	- Completed a thorough reading of the course syllabus and grading criteria - Reviewed the Modern Database Architectures module on Brightspace - Conducted in-depth independent reading on ANN indexing algorithms — Flat, IVF, and HNSW — and the FAISS index types - Reviewed the key literature, including the HNSW paper (Malkov & Yashunin) and the FAISS paper (Johnson et al.)
Outcome	Completed Week 1 with a strong grounding in ANN indexing algorithms, ready to lead the vector indexing and tuning lane.

Week 2 — Team Formation, Role Assignment & Proposal Writing

Actions Taken	- Connected with all three team members — Devu, Rishi, and Nikhil — upon team formation - Participated in the first team meeting to divide responsibilities into four non-overlapping lanes - Took formal ownership of the Vector Indexing & Tuning lane (indexing.py and tune_parameters.py) - Contributed to setting up the shared repository and GitHub Projects board - Began drafting the Related Work and Novelty & Contributions sections of the Pre-Proposal
Outcome	Team roles defined with no overlap; indexing and tuning lane owned; Related Work and Novelty drafting underway.

Week 3 — Pre-Proposal Submission & Index Design

Actions Taken	- Finalized the Related Work and Novelty & Contributions sections of the Pre-Proposal - Designed the unified index abstraction so Flat, IVF, and HNSW could be compared under identical conditions - Planned the FAISS and tuning parameters: IVF nlist / nprobe and HNSW M / efConstruction / efSearch - Participated in full-team review and contributed to the on-time Pre-Proposal submission
Outcome	Pre-Proposal submitted; the indexing methodology and parameter plan were finalized.

Week 4 — Peer Review & FAISS Environment Setup

Actions Taken	• Attended the in-person instructor peer review and defended the choice of ANN indexing techniques • Incorporated instructor feedback on the Related Work into the Full Proposal draft • Set up the FAISS environment and prototyped the common index interface (BaseIndex) • Contributed the index construction / methodology section to the Full Proposal
Outcome	FAISS environment ready; the index abstraction was prototyped; methodology section drafted.

Week 5 — Full Proposal Submitted & Index Scaffolding

Actions Taken	• Finalized the Related Work, Novelty, and indexing methodology in the Full Proposal • Delivered the indexing-methodology portion in the team's recorded Full-Proposal presentation • Committed the initial index scaffolding to the GitHub repository • Supported final proofreading and submission of the Full Proposal
Outcome	Full Proposal submitted and presented; index scaffolding committed to the repository.

Week 6 — Flat Index Implementation & Phase 3 Start

Actions Taken	• Implemented FlatIndex — exhaustive exact search serving as the ground-truth baseline — with memory and build-time measurement • Finalized the FAISS environment and received the embedding batches from the Data & Embeddings Lead (Devu) • Validated that the Flat index returns exact nearest neighbours for use as the recall reference • Updated the GitHub Projects board; Phase 3 index construction formally began
Outcome	Flat index built and serving as the ground-truth baseline; Phase 3 underway.

Week 7 — IVF Index Built & Tested

Actions Taken	- Implemented IVFIndex with k-means partitioning and the nprobe search parameter - Added automatic nlist scaling so clustering stays valid on small datasets - Confirmed Flat and IVF indexes build and query correctly under identical experimental conditions - Supported the Query & Evaluation Lead (Rishi) as initial benchmarking began for Flat and IVF
Outcome	Flat and IVF indexes complete and validated; benchmarking started for both methods.

Week 8 — HNSW Index & Parameter-Tuning Framework (Current)

Actions Taken	- Implementing HNSWIndex — the multi-layer proximity graph with M, efConstruction, and efSearch parameters - Verifying that all three indexes return comparable top-K results under the same conditions - Recording index memory footprint and construction time for each method - Began implementing tune_parameters.py to sweep IVF nprobe and HNSW efSearch (parameter tuning) - Began drafting the index construction / methodology section of the final report
Outcome	HNSW construction in progress and the parameter-tuning framework underway; Flat and IVF complete; Phase 3 progressing into Phase 4.

3. Current Project Status

The project is active and ongoing. Phases 1 and 2 are complete and the team is in Phase 3, with HNSW index construction underway and progressing into Phase 4. The Flat and IVF indexes I built are complete and validated; HNSW construction and the parameter-tuning sweeps are the remaining tasks in my lane.

Phase 1	Dataset Collection & Preprocessing	Complete	ABC News Headlines ingested, cleaned, normalized, and segmented using data_ingestion.py and prepare_headlines.py.
Phase 2	Embedding Generation	Complete	Deterministic 384-dimensional Sentence-BERT embeddings generated via embeddings.py for all corpus sizes.
Phase 3	Index Construction	In Progress	Flat and IVF indexes built and tested; HNSW index currently under construction in FAISS.
Phase 4	Benchmarking & Evaluation	In Progress	Initial benchmarking runs begun for Flat and IVF; HNSW benchmarking to follow as index construction completes.

Phase 5	Analysis & Documentation	Not Started	Report writing, results analysis, and presentation preparation scheduled over the remaining weeks.
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Phase 3 — Index Construction (My Core Phase)

Using the embeddings produced by the data lane, I implemented the FAISS indexes under identical conditions. FlatIndex provides exact search and the recall ground truth; IVFIndex partitions the space with k-means and searches a subset of clusters via nprobe; HNSWIndex builds a multi-layer proximity graph navigated by greedy search. Flat and IVF are complete and tested; HNSW construction is currently underway, after which the memory and build-time figures for all three will be finalized.

Phase 4 — Parameter Tuning (My Experiments)

I also own the parameter-tuning experiments for the indexes. `tune_parameters.py` sweeps IVF nprobe and HNSW efSearch and records recall and latency at each setting, mapping the speed/accuracy trade-off. The framework is being implemented; the full sweeps will run once HNSW construction completes.

Remaining Work

My remaining tasks are to complete HNSW construction, finalize the per-method memory and build-time figures, run the parameter-tuning sweeps, and write the index construction / methodology section of the final report.